Project 2:
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Problem 1

a)

In order for the UCC to be an M/M/1-queue, the following conditions need to be satified:

- Interarrival times are i.i.d $\sim$ Exp(λ) (memoryless)
- Service times are i.i.d $\sim$ Exp(μ) (memoryless)
- There is one "server" and service times are independent of the arrival process.

The arrivals follow a Poisson process with rate $\lambda > 0$, and by thm. 5.5 in the texbook, the sojourn times in between arrivals (interarrival times) are independent and (identically) exponentially distributed with parameter λ. So, the first condition is satisfied.

From the task description, we have that treatment (service) times are i.i.d exponentially distributed with expected value $1/\mu$, and thus i.i.d $\sim$ Exp(μ). Thus, the second condition is satisfied.

Finally, we have one person treating the patients, i.e. one server, and the task description explicitly states that treatment times are independent of arrival process.

So, all conditions are satisfied for the UCC to be modelled as an M/M/1-queue.

The stochastic process $\{X(t) : t \geq 0\}$ can be viewed as a birth and death process, because the possible transitions at each "event" is that X increases by one (if the event is an arrival), or that it decreases by one (if the event is a patient completing its treatment). Also, it is a CTMC with discrete (infinite) state space $S = \{0, 1, 2, ...\}$:

- The probability for a certain transition depends only on the current state, and not on the previous history. Arrivals follow a Poisson process, which has independent increments and the arrival times are memoryless. Treatment times are exponentiallydistributed, hence are also memoryless, and arrivals and treatment times are independent. The Markov property is satisfied.
- The probability of being in one state at time t and another state at time $t + s$ is independent of t, and only depends on the time difference s between the two. We have stationary transition probabilities.
- $S = \{0, 1, 2, ...\}$ since we can have 0 or more patients in the UCC, but there is no limit to the capacity.

Also, the probabilities of an arrival happening in a small time interval h is $\lambda h + o(h)$ since the arrivals follow a Poisson process with rate λ. The departure times for $X(t) \geq 1$ follow an exponential distribution, and the probability of a treatment being completed

during a small time interval h is $P(T \leq h) = 1 - e^{-\lambda h}$ which can be Taylor expanded for small h to $P(T \leq h) \approx \mu h + o(h)$. The probability of remaining in the same state is:

$$P_{i,i}(h) = 1 - P_{i,i+1}(h) - P_{i,i-1}(h) = 1 - \lambda h - \mu h + o(h) = 1 - (\lambda + \mu)h + o(h)$$

. $P_{i,j} = o(h)$ as $h \longrightarrow 0$ since the probability of completing more than one step during a small time interval tends to zero when h is small. Also, $P_{i,j}(0) = \delta_{i,j}$ obviously, and since no departures can happen from state 0, $\mu_0 = 0$. Thus, the conditions for a birth-and-death process are satisfied.

To determine the average time spent in the UCC by a patient, we find the limiting distribution. The limiting distribution exists because $\lambda < \mu$.

We find π_i through examining probability flux balance:

$$\pi_0\lambda = \pi_1\mu, \pi_1\lambda = \pi_2\mu, ...$$

, so

$$\pi_i = \pi_0(\frac{\lambda}{\mu})^i$$

. We find $\pi_0^{-1} = \Sigma_{n=0}^{\infty}(\frac{\lambda}{\mu})^n = \frac{1}{1-\frac{\lambda}{mu}} = \frac{\mu}{\mu-\lambda}$ since $\frac{\lambda}{\mu} < 1$. Thus,

$$\pi_i = \frac{\mu - \lambda}{\mu}(\frac{\lambda}{\mu})^i = (1 - \frac{\lambda}{\mu})(\frac{\lambda}{\mu})^i$$

We recognize the similarity to the geometric distribution, with $f(y) = p(1-p)^{y-1}, y = 1, 2, ...$, with $p = 1 - \frac{\lambda}{\mu}$. However, we have states $0, 1, 2, ...$, so the limiting distribution of $X \sim Y - 1$, and thus $L = E(X) = E(Y - 1) = 1/p - 1 = \frac{\lambda}{\mu-\lambda}$.

Using Little's Law

$$L = \lambda W \tag{1}$$

we find $W = \frac{L}{\lambda} = \frac{1}{\mu-\lambda}$.

b)

We have an arrival rate of $\lambda = 5/\text{hour} = \frac{5}{60}/\text{min}$ and departure rate of $\mu = \frac{1}{10}/\text{min}$. We simulate the process $\{X(t) : t \geq 0\}$ for 50 days. See code attached.

Since all states communicate, the process is irreducible. Since $\lambda < \mu$ it is also positive recurrent, and thus ergodic. Therefore we can approximate the expected time a patient spends in the UCC by estimating the long-run mean number of patients in the UCC, and using 1 to find an estimate for the expected time. This corresponds to considering $E(X*) = \lim_{t\to\infty} \frac{1}{t}(\int_0^t X(t)dt)$. We use a weighted mean $\bar{X} = \frac{\Sigma_{i=1}^N s_i x_i}{\Sigma_{i=1}^N s_i}$ to estimate the long-run mean number of patients in the UCC, where N is the number of events during the 50 days, s_i is time spent in state x_i before a new event occurred. Also, we use only the final half of the array of states to "skip" the burn-in period.

See code.

After one realization we got $L \approx 5$ so $W \approx 60$. We then repeat the simulation 30 times, and by the Central Limit Theorem, we approximate the average of the 30 estimates W_i as normally distributed, and approximate the distribution of the pivotal we get when replacing the unknown σ with empirical standard deviation

$S = \sqrt{\frac{1}{29}\Sigma_{i=1}^{30}(\frac{\bar{W}-\mu_W}{S_W/\sqrt{30}})}$ as $\sim t_{29}$.

We find the 95% confidence interval to be $[\bar{W}-t_{29,0.025}S_W/\sqrt{30}, \bar{W}+t_{29,0.025}S_W/\sqrt{30}]$, with $t_{29,0.025} = 2.045$. One run of the code gave CI $[56.8, 66.5]$ for W. The exact value is $W = 60$ minutes, which is within this interval. In order to verify the CIs being 95% CIs, we generate the CI 1000 times and find the proportion of time where the exact value lies within the computed interval, which was 93.7%. Not exactly 95% but this can perhaps be explained by the normal approximation. See code.

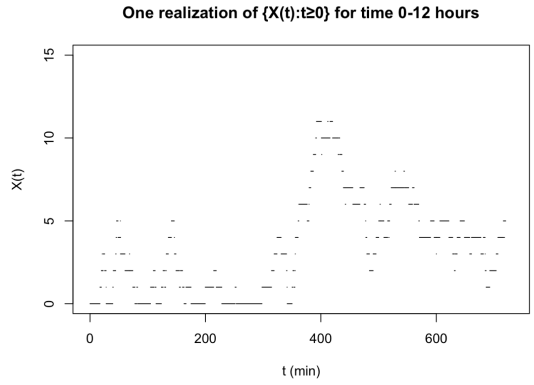


Figure 1: Plot of realizations for 12 hours.

Figure 1 shows one realization of $\{X(t) : t \geq 0\}$ for time $0 - 12$ hours.

c)

$\{U(t) : t \geq 0\}$ satisfies the conditions for an M/M/1-queue because:

- Patients in general arrive with rate λ and are urgent with probability p, so the arrivals of urgent patients follow a Poisson process with rate $p\lambda$. Thus, the inter-arrival times are i.i.d exponentially distributed with parameter $p\lambda$.
- When urgent patients arrive they are treated immediately if there are no other urgent patients in the UCC, and if there is they wait.
- Treatment times are i.i.d $\sim$ Exp(μ) as before.
- As before, we have one server and the service times are independent of arrival process.

$p \in (0,1)$ so $p\lambda < \lambda < \mu$ and thus we have a limiting distribution $\pi_i = (1 - \frac{\lambda p}{\mu})(\frac{\lambda}{\mu})^i$ which by the same reasoning as before yields $L_U = E(U) = \frac{1}{1-\frac{\lambda p}{\mu}} - 1 = \frac{\lambda p}{\mu - \lambda p}$, where L_U is the long-run mean (or expected) number of urgent patients in the UCC.

$\{N(t) : t \geq\}$ does not behave as an M/M/1-queue because:

- Treatment times can be interrupted by an urgent arrival.
- N can only decrease by 1 if $U = 0$, i.e. a normal patient is in treatment, and the treatment is completed before an urgent patient arrives. This happens at rate μ only if $U = 0$. But, N can increase by 1 regardless of the state $U(t)$.
- The server is not always available for normal patients.
- The transition probabilities for N to go from $i \to i - 1$ depends on the state of U and the Markov property is not satisfied! Therefore it cannot be an M/M/1-queue.

Still, we can find the long-run mean number of normal patients in the UCC. $X(t) = N(t) + U(t)$ so $E(N) = E(X) - E(U) = \frac{\lambda}{\mu - \lambda} - \frac{\lambda p}{\mu - \lambda} = \frac{\lambda\mu(1-p)}{(\mu-\lambda)(\mu-\lambda p)}$.

e)

By 1 $W_U = \frac{L_U}{\lambda p} = \frac{1}{\mu - \lambda p}$ and $W_N = \frac{L_N}{(1-p)\lambda} = \frac{\mu}{(\mu-\lambda)(\mu-\lambda p)}$.

f)

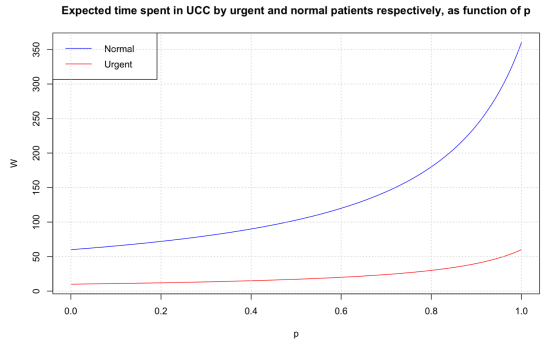


Figure 2: Plot of W_U, W_N as a function of $p \in (0,1)$.

Figure 2 shows the joint plot of W_U, W_N as function of p.

The case of $p \approx 0$ corresponds to almost no arriving patients being urgent, i.e. almost all arriving patients are normal, while the case of $p \approx 1$ corresponds to almost all arriving patients being urgent and thus almost none being normal.

For $p \approx 0$: $\lim_{p\to 0} W_N = \lim_{p\to 0} \frac{\mu}{(\mu-\lambda)(\mu-\lambda p)} = \frac{1}{\mu-\lambda} = 60\text{min} = E(X)$ in the previous exercises. So, in the limit $p \to 0$, $N(t)$ behaves as an M/M/1-queue with all normal patients.

For $p \approx 1$: $\lim_{p\to 1} W_N = \lim_{p\to 0} \frac{\mu}{(\mu-\lambda)(\mu-\lambda p)} = \frac{\mu}{(\mu-\lambda)^2} = 360\text{min}$,i.e. a longer waiting time for normal patients when a lot of arriving patients are urgent (as expected).

In order for W_N to be 2 hours = 120 minutes, $\frac{\mu}{(\mu-\lambda)(\mu-\lambda p)} = 120$ so $6 = 120(\frac{1}{10} - \frac{5}{60}p)$ so $p = \frac{3}{5}$.

g)

Now, $p = 0.80$. We simulate $\{U(t) : t \geq 0\}$ and $\{N(t) : t \geq 0\}$ jointly for 50 days. See code.

The expected times in UCC for urgent and normal patients respectively is $W_U = 30$ min and $W_N = 180$ min.

To find an approximate 95% CI for W_U and W_N we use a normal approximation as before, and find $[\bar{W} - t_{n-1,0.025}S/\sqrt{n}, \bar{W} + t_{n-1,0.025}S/\sqrt{n}]$, with $t_{n-1,0.025} = 2.045$. From one computation of the CIs we got $CI_U = [28.8, 31.3], CI_N = [165.3, 203.4]$ which both contain the true values. Verified by computing 1000 times and found the true values to be contained in the intervals 94.2% of the time for W_U, and 92.4% of the time for W_N. Again, the intervals not being truly 95% CIs can perhaps be explained by it being an approximation.

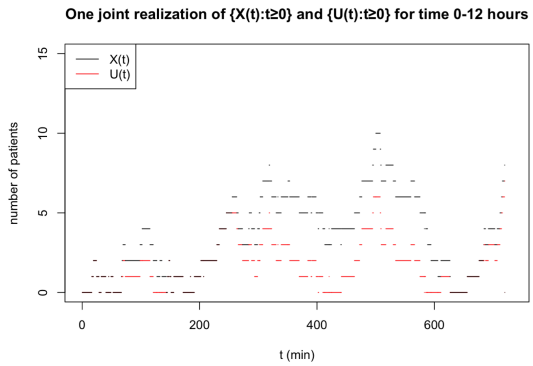


Figure 3: Visualization of $U(t)$ and $X(t) = U(t) + N(t)$ as a function of time.

Figure 3 shows the joint realization of $U(t), X(t)$ as function of time for time $0 - 12$ hours.

Problem 2

a)

A Gaussian process model has a multivariate normal distribution.

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

To compute the conditional mean and the conditional covariance matrix we had to first compute the mean given by:

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu_1} \\ \boldsymbol{\mu_2} \end{pmatrix}$$

In the task description, the expectation value was given by $E[Y(\theta] \equiv 0.5$, and therefore $\boldsymbol{\mu_1} = (0.5, 0.5, ..., 0.5)$ containing 46 elements, and representing the expectation values for the unknown fit of each of the 46 parameters on the regular grid. And $\boldsymbol{\mu_2} = (0.5, 0.5, 0.5, 0.5, 0.5)$, being the respective expectation values for the fit of the given evaluation points.

And then compute the covariance matrices:

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

Here, each element in the different matrices is $\Gamma_{i,j} = Cov[X_i, X_j]$. On the diagonal this will be equal to the variance of the ith element of the grid.

The conditional mean and conditional covariance matrix are given by the following equations.

$$E[\boldsymbol{X_1}|\boldsymbol{X_2} = \boldsymbol{x_2}] = \boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2)$$

$$Cov[\boldsymbol{X_1}|\boldsymbol{X_2} = \boldsymbol{x_2}] = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Every single one of these vectors and matrices was computed by code, and can be found in the attachment.

The next part of the task was to display the prediction as a function of θ, along with the corresponding 90% prediction intervals.

The elements on the diagonal of the conditional covariance matrix are the conditional variance terms. Together with the conditional mean, they define the conditional marginal distribution $p(x_{1,i}|\boldsymbol{x_2}), \quad i = 1, ...n$. So $X_{1,i} \mid \boldsymbol{X_2} = \boldsymbol{x_2} \sim \mathcal{N}(\mathbf{E}(X_{1,i} \mid \boldsymbol{x_2}), \mathrm{Var}(X_{1,i} \mid \boldsymbol{x_2}))$, $Z_i = \frac{X_{1,i}|\boldsymbol{x_2} - \mathbf{E}(X_{1,i}|\boldsymbol{x_2})}{\sqrt{(\mathrm{Var}(X_{1,i}|\boldsymbol{x_2}))}} \sim \mathcal{N}(0, 1)$. The 90% conditional prediction interval at this time point, given $\boldsymbol{x_2}$, is then given by:

$$\left\{ \mathbf{E}(X_{1,i} \mid \boldsymbol{x_2}) \pm z_{0.05}\sqrt{\mathrm{Var}(X_{1,i} \mid \boldsymbol{x_2})} \right\}$$

where $z_{0.05}$ is the upper 5 percent quantile of the standard Gaussian pdf.

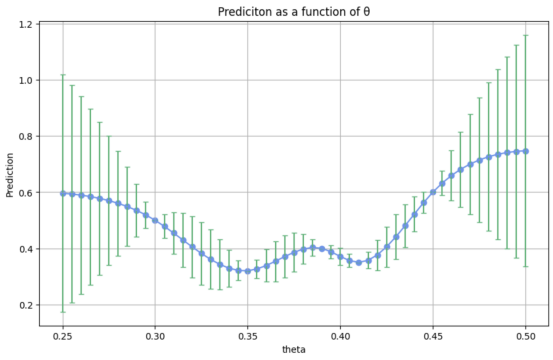


Figure 4: The prediction displayed as a function of θ , with the 90 % prediction intervals

Here in figure 4 we can see by the size of the prediction intervals that the predictions close to the given evaluation points are smaller compared to the θ values further away e.g. on the edges. This shows that the climate model predicts better for the points closer to where the true values are centered.

b)

In task b) the scientist's goal was to achieve $y(\theta) < 0.30$. Together with the prediction from task a) we used the conditional probability given by:

$$p(X_{1,i} < a \mid \boldsymbol{x_2}) = \Phi\left(\frac{a - \mathbf{E}(X_{1,i} \mid \boldsymbol{x_2})}{\sqrt{\mathrm{Var}(X_{1,i} \mid \boldsymbol{x_2})}}\right)$$

where Φ is the cumulative distribution function of the standard Gaussian pdf.

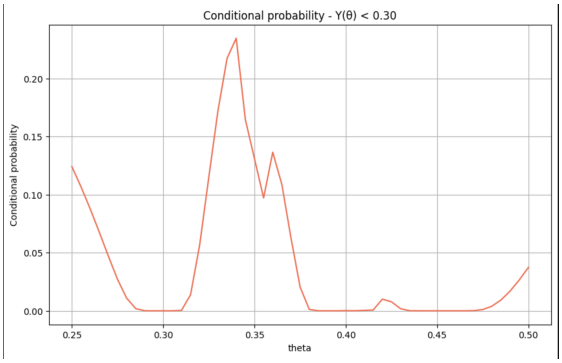
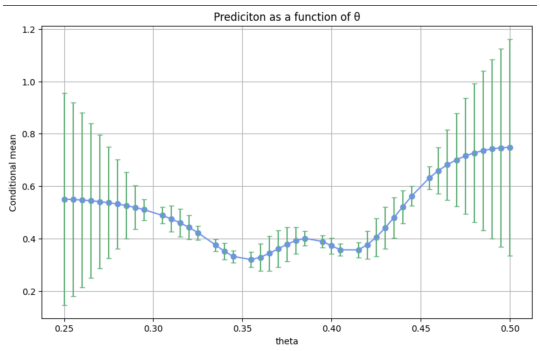


Figure 5: The conditional probability that $Y(\theta) < 0.30$ given the five evaluation points

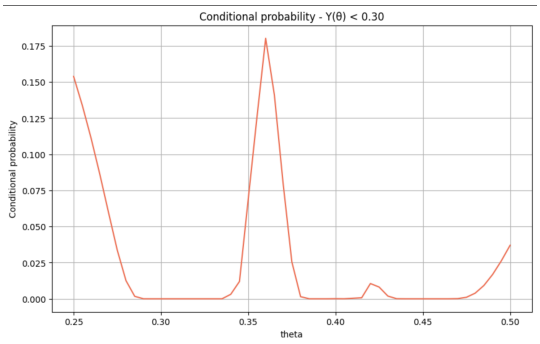
From figure 5 we can see that $\theta = 0.25$ has a probability of 0.125 of being below the threshold of 0.30. And at $\theta = 0.34$ and $\theta = 0.36$ we can see that there are two peaks with corresponding probabilities around 0.24 and 0.135.

c)

In task c) we added the new evaluation point $(\theta, y(\theta) = (0.33, 0.40)$ and reused previous code to compute new conditional means, conditional covariance matrix, predictions, 90% prediction intervals and probabilities $Y(\theta) < 0.30$.



(a) The new prediction after adding the evaluation point $(\theta, y(\theta) = (0.33, 0.40)$



(b) The conditional probability that $Y(\theta) < 0.30$ given the six evaluation points

Figure 6

After adding the new evaluation point the prediction intervals around the center of the considered interval are smaller than before adding the point. This means that the climate model predicts even better now that it has another point to compare to.

The scientist's budget allows for one more run of the climate model, and we would recommend $\theta = 0.36$ since that is the θ-value with the highest conditional probability as well as having a small prediction interval. And will therefore most likely have the best chance to achieve $y(\theta) < 0.30$.

$\theta = 0.25$ also has a high conditional probability, however the prediction interval is considerably larger which makes the prediciton more uncertain. If there was more room in the budget this could be an interesting value to investigate, since the evaluation points are pretty centered, and this would provide information of what happens closer to the start of the interval.